SUPPLY MANAGEMENT SYSTEM — Functional Specification Document

FSD Addendum 41 — CRM Module: Lead Management, Pipeline & Activity Tracking

Property	Value
Document ID	SMS-FSD-ADD-041
Version	1.0
Date	2026-10-11
Status	Draft — Pending Approval
Author	System Architect
Classification	Internal — Development
Supersedes	None — new module under `crm` schema
Architecture	.NET 8 / EF Core 8 / SQL Server 2022 / React 18
Change Items	10 new features
Impact	14 migrations, 12 new entities, 2 modified entities, 3 new lookups, 8 new UI panels, 2 state machines
Depends On	Addendum 37 (Module Registry — CRM module activation), Addendum 32 (Sale Inquiry, Sale Quotation, Sale Order extensions)
Deferred to Addendum 42	Embeddable Lead Widget, External System Sync (Odoo/SAP/Dynamics connectors), Lead Scoring Rules Engine, AI Conversation Intelligence
Naming Convention	New CRM entities follow Microsoft Dynamics 365 naming (Lead, Opportunity, Activity, Campaign). Existing SMS entities retain their names (SaleInquiry, SaleQuotation, SaleOrder, BusinessPartner).

Table of Contents

1. Purpose & Scope
2. CRM Pipeline Overview

3. C1 — CRM Module Activation & Schema
4. C2 — Lead Sources & Campaigns
5. C3 — Sales Teams
6. C4 — Leads
7. C5 — Pipeline Stages
8. C6 — Lead Qualification & Conversion
9. C7 — Opportunities
10. C8 — Activities & Timeline
11. C9 — Activity Documents
12. C10 — Timeline View, Export & Reporting
13. Database Migrations
14. API Endpoints
15. UI Wireframes & Specifications
16. Business Rules
17. Test Scenarios
18. Development Phases

---

1. Purpose & Scope

This addendum introduces the CRM Module — a complete Lead-to-Opportunity pipeline that captures unqualified prospects, tracks all sales conversations, and converts qualified leads into the existing Sales Pre-Order Pipeline (Sale Inquiry → Sale Quotation → Sale Order) defined in Addendum 32.

The CRM module fills the gap before the Sale Inquiry stage. Currently, SMS requires a BusinessPartner (customer) to exist before a Sale Inquiry can be created. This addendum adds a **Lead** entity that holds prospect information without requiring a customer record, and a **qualification process** that creates the BusinessPartner and Opportunity when the lead is ready.

1.1 Design Principles

- **Dynamics 365 Naming & Field Compatibility** — New CRM entities use Microsoft Dynamics 365 naming conventions (Lead, Opportunity, Activity, Campaign) and carry Dynamics-compatible fields to support future data sync with Dynamics, Odoo, and SAP installations.
- **No Premature Customer Creation** — A Lead exists independently of BusinessPartner. Customer and Contact records are created only when the lead is qualified, never before.
- **Individual as Company** — Individual customers (B2C) are managed using the person's name as the company name, following Dynamics' personal account pattern. Every Lead

converts to a BusinessPartner + SupplierContact regardless of whether the prospect represents a company or an individual.

- **Complete Conversation Capture** — Every phone call, meeting, email, note, and document exchange between the salesperson and the prospect is recorded in a unified Activity Timeline. This data feeds future analytics for sales gap analysis and product improvement insights.
- **Document Tracking Per Step** — Every document shared with or received from the customer (brochures, price lists, proposals, customer POs) is tracked with direction, stage context, and timestamp.
- **Multi-Company Contact Isolation** — The same physical person may appear as a contact at multiple companies. Lead data, conversations, and documents for Company A are never visible when viewing the same person's records at Company B. Isolation is enforced by the Account (BusinessPartner) FK on the Lead, not by the contact person record.
- **Cross-Reference Visibility** — When viewing a Lead, the UI shows other Leads for the same Account (company) and the same Contact Person, enabling salespeople to see relationship history and avoid duplicating effort.
- **Configurable Pipeline** — Pipeline stages are configurable per organization and optionally per sales team. Stages can require specific fields or activities before a Lead advances.
- **Timeline Export** — The complete conversation timeline can be viewed in the UI, exported as an Excel workbook, or generated as a formatted PDF report.
- **API-First Lead Ingestion** — Lead creation is exposed via a documented REST API with source attribution, laying the foundation for future embeddable widgets, Odoo/SAP/Dynamics connectors, and third-party integrations (Addendum 42).
- **Multi-Tenant by Default** — Every new entity carries `org_id` with EF Core `HasQueryFilter`. No exception.

1.2 Scope Summary

Change ID	Title	Type	Impact
C1	CRM Module Activation & Schema	Module Config	Activate CRM module in Module Catalog, create `crm` schema
C2	Lead Sources & Campaigns	New Lookups + Entity	2 new tables (LeadSources, Campaigns)
C3	Sales Teams	New Entity	2 new tables (SalesTeams, SalesTeamMembers)
C4	Leads	New Entity + UI	1 new table (Leads), 1 state machine, 2 new UI panels
C5	Pipeline Stages	New Entity	1 new table (PipelineStages) with configurable stages
C6	Lead Qualification & Conversion	Business Logic	Conversion service creating BP + Contact + Opportunity from Lead
C7	Opportunities	New Entity + UI	1 new table (Opportunities), 1 state machine, 1 new UI panel
C8	Activities & Timeline	New Entity + UI	2 new tables (Activities, ActivityParticipants), 1 new UI panel
C9	Activity Documents	New Entity	1 new table (ActivityDocuments)
C10	Timeline View, Export & Reporting	Feature + UI	Timeline panel, Excel export, PDF generation, cross-reference panel

1.3 Relationship to Existing Entities

Existing Entity	Schema	Addendum 41 Interaction
suppliers.BusinessPartners	suppliers	REFERENCED — Lead.account_id FK after qualification; created by qualification service
suppliers.SupplierContacts	suppliers	REFERENCED — Lead.contact_person_id FK after qualification; created by qualification service
demand.SaleInquiries	demand	REFERENCED — Opportunity.source_inquiry_id FK; Inquiry can be created from Opportunity
demand.SaleQuotations	demand	REFERENCED — Opportunity.source_quotation_id FK; Quotation can be created from Opportunity
demand.SaleOrders	demand	UNCHANGED — existing source_type extended with 'FROM_OPPORTUNITY'
core.ModuleCatalog	core	MODIFIED — CRM module row updated: is_available = 1
core.FileStorage	core	REFERENCED — ActivityDocuments.file_id FK
auth.Users	auth	REFERENCED — Lead.assigned_to_user_id, Activity.performed_by_user_id, etc.

1.4 Dynamics 365 Compatibility Matrix

This matrix maps Dynamics 365 Sales entities to SMS equivalents for future data sync.

Dynamics 365 Entity	SMS Entity	Schema	Notes
Lead	crm.Leads	crm	New — this addendum
Account	suppliers.BusinessPartners	suppliers	Existing — `is_customer = true`
Contact	suppliers.SupplierContacts	suppliers	Existing — despite "Supplier" name, holds all BP contacts
Opportunity	crm.Opportunities	crm	New — this addendum
Activity (Phone Call, Appointment, Task, Email, Note)	crm.Activities	crm	New — this addendum
Campaign	crm.Campaigns	crm	New — this addendum
Quote	demand.SaleQuotations	demand	Existing — Addendum 32
Order	demand.SaleOrders	demand	Existing
Invoice	finance.SalesInvoices	finance	Existing
Product	lookups.Products	lookups	Existing

2. CRM Pipeline Overview

2.1 Complete Lead-to-Sale Pipeline

2.2 Document Numbering

Document	Pattern	Example	Auto-Generated
Lead	LD-{YYYY}-{NNNN}	LD-2026-0001	Yes — per org, per year, sequential
Opportunity	OP-{YYYY}-{NNNN}	OP-2026-0001	Yes — per org, per year, sequential
Sale Inquiry	INQ-{YYYY}-{NNNN}	INQ-2026-0001	Yes — existing (Addendum 32)
Sale Quotation	SQ-{YYYY}-{NNNN}	SQ-2026-0001	Yes — existing (Addendum 32)
Sale Order	SO-{NNNN}	SO-0001	Yes — existing

2.3 Creation Paths

The CRM pipeline adds new entry paths while preserving all existing direct-creation paths from Addendum 32.

```
  Path 1: Lead → Opportunity → Inquiry → Quotation → Sale Order  (full CRM chain)
  Path 2: Lead → Opportunity → Quotation → Sale Order            (skip inquiry)
  Path 3: Lead → Opportunity → Sale Order                         (direct from
opportunity)
  Path 4: Inquiry → Quotation → Sale Order                        (existing, no CRM —
Add 32)
  Path 5: Quotation → Sale Order                                  (existing — Add 32)
  Path 6: Sale Order                                               (existing direct
entry)
```

---

3. C1 — CRM Module Activation & Schema

3.1 Overview

The CRM module was registered as a **future** module in the Module Catalog (Addendum 37) with `is_available = 0`. This change activates it and creates the `crm` database schema.

3.2 Module Catalog Update

Update the existing seed data row:

```sql
UPDATE core.ModuleCatalog
SET is_available = 1,
    display_name = 'CRM — Lead & Opportunity Management',
    description = 'Lead capture, qualification, pipeline tracking, and conversion to sales
WHERE module_code = 'CRM';
```

3.3 Module Features

Feature Code	Display Name	Is Core	Is Available	Description
LEAD_MANAGEMENT	Lead Management	Yes (core)	1	Lead capture, qualification, conversion
OPPORTUNITY_MANAGEMENT	Opportunity Management	Yes (core)	1	Deal tracking, pipeline stages, revenue forecasting
ACTIVITY_TRACKING	Activity & Timeline	Yes (core)	1	Conversation capture, meeting notes, document tracking
CAMPAIGN_TRACKING	Campaign Tracking	No (toggleable)	1	Lead source attribution, campaign management
LEAD_ASSIGNMENT	Lead Assignment Rules	No (toggleable)	0	Automatic lead routing (Phase 2)
LEAD_SCORING	Lead Scoring	No (toggleable)	0	Rule-based lead scoring (deferred to Addendum 42)
LEAD_WIDGET	Embeddable Lead Widget	No (toggleable)	0	External lead capture forms (deferred to Addendum 42)
EXTERNAL_SYNC	External CRM Sync	No (toggleable)	0	Odoo/SAP/Dynamics sync (deferred to Addendum 42)

3.4 Schema Creation

```sql
-- M1: Create CRM schema
IF NOT EXISTS (SELECT 1 FROM sys.schemas WHERE name = 'crm')
    EXEC('CREATE SCHEMA crm');
```

3.5 CRM Module Dependency

CRM depends on CUSTOMERS (always-on) per Addendum 37 dependency graph. No additional dependencies are introduced beyond what the Module Catalog already defines.

---

4. C2 — Lead Sources & Campaigns

4.1 Overview

Lead Sources are a configurable lookup defining how leads enter the system (Facebook, WhatsApp, website, walk-in, etc.). Campaigns are specific marketing initiatives that generate leads and carry attribution data.

4.2 Schema — LeadSources

New Table: `crm.LeadSources`

This is a tenant-configurable lookup. SMS ships seed data; organizations can add their own.

Column	Type	Nullable	Default	Description
lead_source_id	INT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
source_code	NVARCHAR(30)	NOT NULL		Machine-readable code: WEB_FORM, FACEBOOK, WHATSAPP, PHONE_CALL, WALK_IN, REFERRAL, EXISTING_CUSTOMER, IMPORT, CAMPAIGN, PARTNER, OTHER
source_name	NVARCHAR(100)	NOT NULL		Display name
description	NVARCHAR(500)	NULL		
is_active	BIT	NOT NULL	1	Soft deactivation
sort_order	INT	NOT NULL	0	Display sequence
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
UQ_LeadSources_OrgCode	org_id, source_code	Unique

Seed Data (per-org, inserted on CRM module activation):

source_code	source_name
WEB_FORM	Website Form
FACEBOOK	Facebook
INSTAGRAM	Instagram
WHATSAPP	WhatsApp
PHONE_CALL	Phone Call
WALK_IN	Walk-in
REFERRAL	Referral
EXISTING_CUSTOMER	Existing Customer
EMAIL	Email
LINKEDIN	LinkedIn
IMPORT	Data Import
PARTNER	Partner/Reseller
TRADE_SHOW	Trade Show / Exhibition
OTHER	Other

4.3 Schema — Campaigns

New Table: `crm.Campaigns`

Column	Type	Nullable	Default	Description
campaign_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
campaign_code	NVARCHAR(30)	NOT NULL		Unique code per org
campaign_name	NVARCHAR(200)	NOT NULL		Display name
description	NVARCHAR(2000)	NULL		Campaign details
campaign_type	NVARCHAR(30)	NOT NULL	'GENERAL'	GENERAL, ADVERTISEMENT, PROMOTION, EVENT, EMAIL_BLAST, SOCIAL_MEDIA, REFERRAL_PROGRAM
lead_source_id	INT	NULL	FK → LeadSources	Default source for leads from this campaign
status	NVARCHAR(20)	NOT NULL	'DRAFT'	DRAFT, ACTIVE, PAUSED, COMPLETED, CANCELLED
start_date	DATE	NULL		Campaign start
end_date	DATE	NULL		Campaign end
budget_amount	DECIMAL(18,2)	NULL		Planned budget
currency_code	NVARCHAR(3)	NULL	FK → currencies	Budget currency
expected_leads	INT	NULL		Target lead count
actual_leads	INT	NOT NULL	0	Running count, incremented on lead creation

Column	Type	Nullable	Default	Description
external_reference	NVARCHAR(200)	NULL		External campaign ID (Facebook Campaign ID, Google Ads ID, etc.)
notes	NVARCHAR(2000)	NULL		
created_by	BIGINT	NOT NULL	FK → auth.Users	
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
is_deleted	BIT	NOT NULL	0	Soft delete

Indexes:

Index	Columns	Type
UQ_Campaigns_OrgCode	org_id, campaign_code	Unique
IX_Campaigns_Status	org_id, status	Non-unique, filtered
IX_Campaigns_Dates	org_id, start_date, end_date	Non-unique

---

5. C3 — Sales Teams

5.1 Overview

Sales Teams group salespeople for lead assignment, pipeline management, and performance tracking. A salesperson can belong to multiple teams. Each team can have its own pipeline stages (optional — falls back to org-level stages if not configured).

5.2 Schema — SalesTeams

New Table: `crm.SalesTeams`

Column	Type	Nullable	Default	Description
sales_team_id	INT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
team_code	NVARCHAR(30)	NOT NULL		Unique code per org
team_name	NVARCHAR(100)	NOT NULL		Display name
description	NVARCHAR(500)	NULL		
team_leader_user_id	BIGINT	NULL	FK → auth.Users	Team lead / manager
is_active	BIT	NOT NULL	1	
default_campaign_id	BIGINT	NULL	FK → Campaigns	Default campaign for leads created by this team
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
UQ_SalesTeams_OrgCode	org_id, team_code	Unique

5.3 Schema — SalesTeamMembers

New Table: `crm.SalesTeamMembers`

Column	Type	Nullable	Default	Description
sales_team_member_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
sales_team_id	INT	NOT NULL	FK → SalesTeams	Team reference
user_id	BIGINT	NOT NULL	FK → auth.Users	Salesperson
role	NVARCHAR(20)	NOT NULL	'MEMBER'	LEADER, MEMBER
is_active	BIT	NOT NULL	1	
joined_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
UQ_SalesTeamMembers_TeamUser	org_id, sales_team_id, user_id	Unique
IX_SalesTeamMembers_User	user_id	Non-unique

---

6. C4 — Leads

6.1 Overview

A Lead is an unqualified prospect — someone who has expressed interest but has not yet been evaluated by the sales team. The Lead entity captures contact information, source attribution, and interest details WITHOUT requiring a BusinessPartner (customer) record to exist.

Leads follow Microsoft Dynamics 365 field naming and carry Dynamics-compatible qualification fields (BANT: Budget, Authority, Need, Timeline).

> 📝 **NOTE:** For individual customers (B2C), the person's full name is stored in `company_name`. This follows the Dynamics personal-account pattern — every Lead has a company name,

> even if the company is the individual themselves.

6.2 Schema — Leads

New Table: `crm.Leads`

Column	Type	Nullable	Default	Description
lead_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
lead_number	NVARCHAR(20)	NOT NULL	Auto	Pattern: LD-{YYYY}-{NNNN}, unique per org
— Topic —				
topic	NVARCHAR(200)	NOT NULL		Short subject (e.g. "AC + Ovens inquiry from Facebook")
description	NVARCHAR(4000)	NULL		Detailed requirement / interest
— Contact Info (captured before BP exists) —				
salutation	NVARCHAR(10)	NULL		Mr, Mrs, Ms, Dr, etc.
first_name	NVARCHAR(100)	NOT NULL		Contact person first name
last_name	NVARCHAR(100)	NOT NULL		Contact person last name
full_name	NVARCHAR(200)	NOT NULL	Computed	first_name + ' ' + last_name
job_title	NVARCHAR(100)	NULL		Contact's job title / designation

Column	Type	Nullable	Default	Description
email	NVARCHAR(200)	NULL		Primary email
phone	NVARCHAR(30)	NULL		Business phone
mobile_phone	NVARCHAR(30)	NULL		Mobile phone
company_name	NVARCHAR(200)	NOT NULL		Company name. For B2C individuals, use the person's full name.
website	NVARCHAR(500)	NULL		Company website
address_line1	NVARCHAR(200)	NULL		
address_line2	NVARCHAR(200)	NULL		
city	NVARCHAR(100)	NULL		
state_province	NVARCHAR(100)	NULL		
postal_code	NVARCHAR(20)	NULL		
country	NVARCHAR(100)	NULL		
— Linked Records (populated during/after qualification) —				
account_id	BIGINT	NULL	FK → BusinessPartners	Linked company (set on qualify or manual link)
contact_person_id	BIGINT	NULL	FK → SupplierContacts	Linked contact person (set on qualify or manual link)
— Source Attribution —				

Column	Type	Nullable	Default	Description
lead_source_id	INT	NULL	FK → LeadSources	How the lead found us
campaign_id	BIGINT	NULL	FK → Campaigns	Which campaign generated this lead
source_medium	NVARCHAR(50)	NULL		UTM medium (social, email, cpc, organic)
source_reference	NVARCHAR(200)	NULL		Ad ID, form ID, referrer name, UTM content
external_source_system	NVARCHAR(30)	NULL		Source system for synced leads: ODOO, SAP, DYNAMICS, HUBSPOT, OTHER
external_source_id	NVARCHAR(200)	NULL		Original record ID in the source system
— Qualification (Dynamics BANT) —				
budget_amount	DECIMAL(18,2)	NULL		Customer's stated budget
budget_currency_code	NVARCHAR(3)	NULL	FK → currencies	
budget_confirmed	BIT	NOT NULL	0	Budget discussed and confirmed?
authority_confirmed	BIT	NOT NULL	0	Decision maker identified?

Column	Type	Nullable	Default	Description
need_identified	BIT	NOT NULL	0	Genuine need established?
purchase_timeframe	NVARCHAR(20)	NULL		IMMEDIATE, THIS_QUARTER, THIS_YEAR, NEXT_YEAR, UNKNOWN
— Rating & Priority —				
rating	NVARCHAR(10)	NOT NULL	'WARM'	HOT, WARM, COLD
priority	NVARCHAR(10)	NOT NULL	'NORMAL'	HIGH, NORMAL, LOW
— Ownership —				
assigned_to_user_id	BIGINT	NULL	FK → auth.Users	Assigned salesperson
assigned_at	DATETIME2	NULL		When assignment was made
sales_team_id	INT	NULL	FK → SalesTeams	Assigned team
previous_owner_user_id	BIGINT	NULL	FK → auth.Users	Last owner before reassignment
— Pipeline —				
pipeline_stage_id	INT	NULL	FK → PipelineStages	Current stage in configurable pipeline
— Status —				

Column	Type	Nullable	Default	Description
status	NVARCHAR(20)	NOT NULL	'NEW'	State machine: NEW → CONTACTED → QUALIFIED → DISQUALIFIED
status_reason	NVARCHAR(500)	NULL		Reason (especially for disqualification)
qualified_at	DATETIME2	NULL		Timestamp of qualification
disqualified_at	DATETIME2	NULL		Timestamp of disqualification
converted_at	DATETIME2	NULL		Timestamp of conversion to Opportunity
— Conversion References —				
converted_account_id	BIGINT	NULL	FK → BusinessPartners	BP created/linked during conversion
converted_contact_id	BIGINT	NULL	FK → SupplierContacts	Contact created/linked during conversion
converted_opportunity_id	BIGINT	NULL	FK → Opportunities	Opportunity created during conversion
— Estimated Deal Value —				

Column	Type	Nullable	Default	Description
estimated_revenue	DECIMAL(18,2)	NULL		Expected deal value
estimated_close_date	DATE	NULL		Expected close date
currency_code	NVARCHAR(3)	NULL	FK → currencies	Revenue currency
— Standard Audit —				
created_by_user_id	BIGINT	NOT NULL	FK → auth.Users	
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_by_user_id	BIGINT	NOT NULL	FK → auth.Users	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
is_deleted	BIT	NOT NULL	0	Soft delete
trace_id	UNIQUEIDENTIFIER	NOT NULL	NEWID()	Correlation ID

Indexes:

Index	Columns	Type
UQ_Leads_OrgNumber	org_id, lead_number	Unique
IX_Leads_Status	org_id, status	Non-unique, filtered (WHERE is_deleted = 0)
IX_Leads_AssignedTo	assigned_to_user_id	Non-unique, filtered (WHERE is_deleted = 0)
IX_Leads_SalesTeam	org_id, sales_team_id	Non-unique, filtered (WHERE is_deleted = 0)
IX_Leads_Campaign	campaign_id	Non-unique
IX_Leads_Account	account_id	Non-unique, filtered (WHERE account_id IS NOT NULL)
IX_Leads_ContactPerson	contact_person_id	Non-unique, filtered (WHERE contact_person_id IS NOT NULL)
IX_Leads_Email	org_id, email	Non-unique, filtered (WHERE email IS NOT NULL) — duplicate detection
IX_Leads_Phone	org_id, mobile_phone	Non-unique, filtered (WHERE mobile_phone IS NOT NULL) — duplicate detection
IX_Leads_CompanyName	org_id, company_name	Non-unique — duplicate detection
IX_Leads_CreatedAt	org_id, created_at DESC	Non-unique — list view sorting
IX_Leads_PipelineStage	org_id, pipeline_stage_id	Non-unique, filtered (WHERE is_deleted = 0) — Kanban grouping

6.3 Lead State Machine

```
Terminal state.
```

Transition Rules:

From	To	Condition	Side Effects
NEW	CONTACTED	At least one Activity (call, meeting, email) recorded	Sets first_contact timestamp
NEW	DISQUALIFIED	Manual action by salesperson or manager	Records status_reason, disqualified_at
CONTACTED	QUALIFIED	Manual action; at least one BANT field confirmed recommended but not enforced	Records qualified_at
CONTACTED	DISQUALIFIED	Manual action	Records status_reason, disqualified_at
QUALIFIED	CONVERTED	Qualification service invoked (C6)	Creates BP + Contact + Opportunity; records converted_at, converted_*_id fields
QUALIFIED	DISQUALIFIED	Manual action (deal fell through before conversion)	Records status_reason, disqualified_at

Re-opening: A DISQUALIFIED lead can be re-opened to NEW status by a user with `crm_reopen_lead` permission. A CONVERTED lead cannot be re-opened — the Opportunity is the active record.

---

7. C5 — Pipeline Stages

7.1 Overview

Pipeline Stages provide a configurable Kanban-style view of leads progressing through the sales process. Stages are configured per organization and optionally per sales team. If a team has no custom stages, it uses the organization's default stages.

7.2 Schema — PipelineStages

New Table: `crm.PipelineStages`

Column	Type	Nullable	Default	Description
pipeline_stage_id	INT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
stage_code	NVARCHAR(30)	NOT NULL		Machine-readable code
stage_name	NVARCHAR(100)	NOT NULL		Display name
stage_type	NVARCHAR(20)	NOT NULL	'LEAD'	LEAD (pre-qualification stages) or OPPORTUNITY (post-qualification stages)
sales_team_id	INT	NULL	FK → SalesTeams	NULL = org-wide default; set = team-specific
sequence	INT	NOT NULL	0	Display order
win_probability	INT	NULL		0–100, displayed on pipeline view
is_won	BIT	NOT NULL	0	Marks the "Won" terminal stage
is_lost	BIT	NOT NULL	0	Marks the "Lost" terminal stage
description	NVARCHAR(500)	NULL		
is_active	BIT	NOT NULL	1	
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	

Indexes:

Index	Columns	Type
UQ_PipelineStages_OrgTeamCode	org_id, sales_team_id, stage_code	Unique
IX_PipelineStages_Sequence	org_id, sales_team_id, stage_type, sequence	Non-unique

Seed Data (per-org, inserted on CRM module activation):

stage_code	stage_name	stage_type	sequence	win_probability
NEW_LEAD	New Lead	LEAD	10	5
INITIAL_CONTACT	Initial Contact	LEAD	20	10
QUALIFICATION	Qualification	LEAD	30	20
NEEDS_ANALYSIS	Needs Analysis	OPPORTUNITY	40	40
PROPOSAL	Proposal / Quotation	OPPORTUNITY	50	60
NEGOTIATION	Negotiation	OPPORTUNITY	60	75
WON	Won	OPPORTUNITY	70	100
LOST	Lost	OPPORTUNITY	80	0

---

8. C6 — Lead Qualification & Conversion

8.1 Overview

When a Lead is qualified, the system creates (or links to existing) BusinessPartner and SupplierContact records, then creates an Opportunity in the CRM pipeline. This is the bridge between the CRM module and the existing SCM pipeline.

8.2 Conversion Process

Service: `ILeadConversionService.ConvertAsync(long leadId, LeadConversionOptions options)`

Pseudocode:

```csharp
```

```
public async Task<LeadConversionResult> ConvertAsync(long leadId, LeadConversionOptions opt
{
    var lead = await _leadRepository.GetByIdAsync(leadId);

    // 1. Validate lead status
    if (lead.Status != LeadStatus.QUALIFIED)
        throw new BusinessRuleException("Only QUALIFIED leads can be converted.");

    // 2. Create or Link Account (BusinessPartner)
    BusinessPartner account;
    if (options.ExistingAccountId.HasValue)
    {
        // Link to existing BusinessPartner
        account = await _bpRepository.GetByIdAsync(options.ExistingAccountId.Value);
        if (!account.IsCustomer)
        {
            account.IsCustomer = true;
            await _bpRepository.UpdateAsync(account);
        }
    }
    else
    {
        // Create new BusinessPartner
        account = new BusinessPartner
        {
            OrgId = lead.OrgId,
            PartnerName = lead.CompanyName,
            IsCustomer = true,
            IsVendor = false,
            Phone = lead.Phone,
            Email = lead.Email,
            Website = lead.Website,
            AddressLine1 = lead.AddressLine1,
            City = lead.City,
            StateProvince = lead.StateProvince,
            PostalCode = lead.PostalCode,
            Country = lead.Country,
            // ... other fields
        };
        await _bpRepository.CreateAsync(account);
    }
```

```
// 3. Create or Link Contact Person (SupplierContact)
SupplierContact contact;
if (options.ExistingContactId.HasValue)
{
    contact = await _contactRepository.GetByIdAsync(options.ExistingContactId.Value);
}
else
{
    contact = new SupplierContact
    {
        OrgId = lead.OrgId,
        BusinessPartnerId = account.BusinessPartnerId,
        Salutation = lead.Salutation,
        FirstName = lead.FirstName,
        LastName = lead.LastName,
        JobTitle = lead.JobTitle,
        Email = lead.Email,
        Phone = lead.Phone,
        MobilePhone = lead.MobilePhone,
        // ... other fields
    };
    await _contactRepository.CreateAsync(contact);
}

// 4. Create Opportunity
var opportunity = new Opportunity
{
    OrgId = lead.OrgId,
    OpportunityNumber = await _numberingService.NextAsync("OP", lead.OrgId),
    Topic = lead.Topic,
    Description = lead.Description,
    AccountId = account.BusinessPartnerId,
    ContactPersonId = contact.SupplierContactId,
    SourceLeadId = lead.LeadId,
    EstimatedRevenue = lead.EstimatedRevenue,
    EstimatedCloseDate = lead.EstimatedCloseDate,
    CurrencyCode = lead.CurrencyCode,
    SalesTeamId = lead.SalesTeamId,
    AssignedToUserId = lead.AssignedToUserId,
    PipelineStageId = _defaultOpportunityStageId, // First OPPORTUNITY-type stage
    Status = OpportunityStatus.OPEN,
    // ... audit fields
```

```
        };
        await _opportunityRepository.CreateAsync(opportunity);

        // 5. Update Lead to CONVERTED
        lead.Status = LeadStatus.CONVERTED;
        lead.ConvertedAt = DateTime.UtcNow;
        lead.ConvertedAccountId = account.BusinessPartnerId;
        lead.ConvertedContactId = contact.SupplierContactId;
        lead.ConvertedOpportunityId = opportunity.OpportunityId;
        lead.AccountId = account.BusinessPartnerId;
        lead.ContactPersonId = contact.SupplierContactId;
        await _leadRepository.UpdateAsync(lead);

        // 6. Copy Activities from Lead to Opportunity
        // Activities remain linked to the Lead but are ALSO visible on the Opportunity
        // via the source_lead_id relationship. No duplication needed.

        // 7. Publish MediatR event
        await _mediator.Publish(new LeadConvertedEvent
        {
            LeadId = lead.LeadId,
            OpportunityId = opportunity.OpportunityId,
            AccountId = account.BusinessPartnerId,
            ContactId = contact.SupplierContactId,
            OrgId = lead.OrgId
        });

        return new LeadConversionResult
        {
            OpportunityId = opportunity.OpportunityId,
            AccountId = account.BusinessPartnerId,
            ContactId = contact.SupplierContactId,
            IsNewAccount = !options.ExistingAccountId.HasValue,
            IsNewContact = !options.ExistingContactId.HasValue
        };
    }
```

8.3 Conversion Options

The conversion UI presents the salesperson with these choices:

Option	Behavior
Create new Account	Creates a new BusinessPartner from Lead's company_name and address fields
Link to existing Account	Salesperson searches and selects an existing BusinessPartner. System shows potential matches based on company_name similarity.
Create new Contact	Creates a new SupplierContact under the selected/created Account
Link to existing Contact	Salesperson selects an existing SupplierContact under the selected Account. System shows contacts for the selected Account only.

8.4 Duplicate Detection During Conversion

Before creating a new Account, the system checks for potential duplicates:

```sql
-- Potential duplicate check
SELECT TOP 5 business_partner_id, partner_name, phone, email
FROM suppliers.BusinessPartners
WHERE org_id = @OrgId
  AND is_deleted = 0
  AND (
      partner_name LIKE '%' + @CompanyName + '%'
      OR email = @Email
      OR phone = @Phone
  )
ORDER BY
    CASE WHEN partner_name = @CompanyName THEN 0 ELSE 1 END,
    partner_name;
```

If duplicates are found, the UI shows them and asks the salesperson to choose: create new or link existing.

8.5 Data Isolation Rule

CRITICAL: When the same physical person (e.g., Ahmed Khan) is a contact at Company A and Company B:

- Each company has its own SupplierContact row (different `contact_id`, same name, different `business_partner_id`)

- Leads for Company A use Company A's `contact_id`
- Leads for Company B use Company B's `contact_id`
- When viewing Ahmed's leads at Company A, the query filters by `account_id = CompanyA_Id`
- Company B's leads, conversations, and documents are NEVER returned

The isolation is enforced at the Account (BusinessPartner) level, not the Contact Person level. The Account FK on the Lead is the primary filter, with Contact as a secondary scope.

---

9. C7 — Opportunities

9.1 Overview

An Opportunity represents a qualified deal — the bridge between the CRM Lead and the existing SCM Sale Inquiry / Quotation pipeline. Opportunities are always created by the Lead Conversion process (C6); they cannot be created standalone.

An Opportunity tracks the deal through pipeline stages, estimates revenue and close date, and links to the Sale Inquiry, Quotation, and/or Sale Order that the deal produces.

9.2 Schema — Opportunities

New Table: `crm.Opportunities`

Column	Type	Nullable	Default	Description
opportunity_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
opportunity_number	NVARCHAR(20)	NOT NULL	Auto	Pattern: OP-{YYYY}-{NNNN}, unique per org
— Topic —				
topic	NVARCHAR(200)	NOT NULL		Deal subject (carried from Lead)
description	NVARCHAR(4000)	NULL		Detailed requirement
— Account & Contact —				
account_id	BIGINT	NOT NULL	FK → BusinessPartners	Company (must exist — created during Lead conversion)
contact_person_id	BIGINT	NULL	FK → SupplierContacts	Primary contact for this deal
— Source —				
source_lead_id	BIGINT	NOT NULL	FK → Leads	Lead that generated this Opportunity
campaign_id	BIGINT	NULL	FK → Campaigns	Carried from Lead
— Pipeline —				

Column	Type	Nullable	Default	Description
pipeline_stage_id	INT	NULL	FK → PipelineStages	Current stage (OPPORTUNITY-type stages)
status	NVARCHAR(20)	NOT NULL	'OPEN'	OPEN, WON, LOST
— Revenue —				
estimated_revenue	DECIMAL(18,2)	NULL		Expected deal value
actual_revenue	DECIMAL(18,2)	NULL		Actual value when won (from SO total)
currency_code	NVARCHAR(3)	NULL	FK → currencies	
estimated_close_date	DATE	NULL		Expected close date
actual_close_date	DATE	NULL		Date when won or lost
— Win/Loss —				
win_loss_reason	NVARCHAR(500)	NULL		Why won or lost
competitor_name	NVARCHAR(200)	NULL		Lost to whom (if applicable)
— Ownership —				
assigned_to_user_id	BIGINT	NULL	FK → auth.Users	Salesperson owning this deal
sales_team_id	INT	NULL	FK → SalesTeams	Team
— Linked SCM Documents —				
source_inquiry_id	BIGINT	NULL	FK → SaleInquiries	Sale Inquiry created from this

Column	Type	Nullable	Default	Description
				opportunity
source_quotation_id	BIGINT	NULL	FK → SaleQuotations	Sale Quotation created from this opportunity
source_order_id	BIGINT	NULL	FK → SaleOrders	Sale Order created from this opportunity
— Standard Audit —				
created_by_user_id	BIGINT	NOT NULL	FK → auth.Users	
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_by_user_id	BIGINT	NOT NULL	FK → auth.Users	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
is_deleted	BIT	NOT NULL	0	Soft delete
trace_id	UNIQUEIDENTIFIER	NOT NULL	NEWID()	

Indexes:

Index	Columns	Type
UQ_Opportunities_OrgNumber	org_id, opportunity_number	Unique
IX_Opportunities_Account	account_id	Non-unique
IX_Opportunities_Contact	contact_person_id	Non-unique, filtered
IX_Opportunities_SourceLead	source_lead_id	Non-unique
IX_Opportunities_Status	org_id, status	Non-unique, filtered
IX_Opportunities_PipelineStage	org_id, pipeline_stage_id	Non-unique, filtered
IX_Opportunities_AssignedTo	assigned_to_user_id	Non-unique, filtered
IX_Opportunities_CloseDate	org_id, estimated_close_date	Non-unique

9.3 Opportunity State Machine

Transition Rules:

From	To	Condition	Side Effects
OPEN	WON	Sale Order confirmed (status = CONFIRMED) linked to this opportunity	Sets actual_close_date, actual_revenue (from SO total), moves pipeline_stage to won-stage
OPEN	LOST	Manual action by salesperson	Requires win_loss_reason. Sets actual_close_date. Moves pipeline_stage to lost-stage
LOST	OPEN	Re-open by user with `crm_reopen_opportunity` permission	Clears actual_close_date, win_loss_reason, competitor_name

Opportunity → SCM Document Creation:

From an open Opportunity, the salesperson can:

Action	Creates	Link
"Create Inquiry"	demand.SaleInquiries	Sets opportunity.source_inquiry_id; inquiry.customer_id = opportunity.account_id
"Create Quotation"	demand.SaleQuotations	Sets opportunity.source_quotation_id; quotation.customer_id = opportunity.account_id
"Create Sale Order"	demand.SaleOrders	Sets opportunity.source_order_id; order.source_type = 'FROM_OPPORTUNITY'

These actions use the existing Addendum 32 creation logic. The only additions are the source reference FKs back to the Opportunity.

9.4 Sale Order Extension

Modified Table: `demand.SaleOrders`

Add one new value to the `source_type` check constraint:

```sql
```

```
-- Existing values: MANUAL, FROM_QUOTATION, PORTAL, INTER_TENANT
-- Add: FROM_OPPORTUNITY
ALTER TABLE demand.SaleOrders
    DROP CONSTRAINT CK_SaleOrders_SourceType;

ALTER TABLE demand.SaleOrders
    ADD CONSTRAINT CK_SaleOrders_SourceType
    CHECK (source_type IN ('MANUAL', 'FROM_QUOTATION', 'PORTAL', 'INTER_TENANT', 'FROM_OPPO
```

Add optional FK:

Column	Type	Nullable	Description
source_opportunity_id	BIGINT	NULL	FK → crm.Opportunities — set when SO is created from an Opportunity

---

10. C8 — Activities & Timeline

10.1 Overview

Activities capture every interaction between the salesperson and the prospect/customer. All activity types are stored in a single table with a `activity_type` discriminator, providing a unified timeline across the entire CRM lifecycle.

Activities are linked to a Lead and/or an Opportunity. When a Lead is converted, its activities remain linked to the Lead but are ALSO visible on the Opportunity through the Lead→Opportunity relationship. New activities after conversion are linked directly to the Opportunity.

10.2 Schema — Activities

New Table: `crm.Activities`

Column	Type	Nullable	Default	Description
activity_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
— Parent Reference —				
lead_id	BIGINT	NULL	FK → Leads	Linked lead (set for pre-conversion activities)
opportunity_id	BIGINT	NULL	FK → Opportunities	Linked opportunity (set for post-conversion activities)
— Activity Type —				
activity_type	NVARCHAR(30)	NOT NULL		PHONE_CALL, MEETING, EMAIL_SENT, EMAIL_RECEIVED, NOTE, TASK, FOLLOW_UP, SITE_VISIT, DEMO, PRESENTATION, STATUS_CHANGE, DOCUMENT_SHARED, OTHER
— Content —				
subject	NVARCHAR(200)	NOT NULL		Short title
description	NVARCHAR(MAX)	NULL		Full conversation notes / meeting minutes / email body

Column	Type	Nullable	Default	Description
outcome	NVARCHAR(30)	NULL		SUCCESSFUL, UNSUCCESSFUL, NO_RESPONSE, FOLLOW_UP_NEEDED, REFERRED, COMPLETED, CANCELLED
outcome_notes	NVARCHAR(2000)	NULL		
— Timing —				
activity_date	DATETIME2	NOT NULL	SYSUTCDATETIME()	When the activity happened
duration_minutes	INT	NULL		Call/meeting duration
— Scheduling —				
scheduled_date	DATETIME2	NULL		If this was a planned activity
due_date	DATETIME2	NULL		Deadline for tasks
completed_date	DATETIME2	NULL		When marked complete
is_completed	BIT	NOT NULL	0	
— Participants —				
performed_by_user_id	BIGINT	NOT NULL	FK → auth.Users	Salesperson who performed the activity
contact_person_id	BIGINT	NULL	FK → SupplierContacts	Customer contact involved
— Follow-up —				

Column	Type	Nullable	Default	Description
follow_up_date	DATE	NULL		Next action due date
follow_up_type	NVARCHAR(30)	NULL		Same enum as activity_type
follow_up_notes	NVARCHAR(500)	NULL		What to do next
— Context —				
pipeline_stage_id	INT	NULL	FK → PipelineStages	Which stage the lead/opportunity was in
— Standard Audit —				
created_by_user_id	BIGINT	NOT NULL	FK → auth.Users	
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
updated_by_user_id	BIGINT	NOT NULL	FK → auth.Users	
updated_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
is_deleted	BIT	NOT NULL	0	Soft delete

Constraint: At least one of `lead_id` or `opportunity_id` must be non-null:

```sql
ALTER TABLE crm.Activities
    ADD CONSTRAINT CK_Activities_ParentRef
    CHECK (lead_id IS NOT NULL OR opportunity_id IS NOT NULL);
```

Indexes:

Index	Columns	Type
IX_Activities_Lead	lead_id, activity_date DESC	Non-unique, filtered (WHERE lead_id IS NOT NULL)
IX_Activities_Opportunity	opportunity_id, activity_date DESC	Non-unique, filtered (WHERE opportunity_id IS NOT NULL)
IX_Activities_PerformedBy	performed_by_user_id, activity_date DESC	Non-unique
IX_Activities_Type	org_id, activity_type, activity_date DESC	Non-unique
IX_Activities_FollowUp	org_id, follow_up_date	Non-unique, filtered (WHERE follow_up_date IS NOT NULL AND is_completed = 0)
IX_Activities_DueDate	org_id, due_date	Non-unique, filtered (WHERE due_date IS NOT NULL AND is_completed = 0)
IX_Activities_Scheduled	org_id, scheduled_date	Non-unique, filtered (WHERE scheduled_date IS NOT NULL AND is_completed = 0)

10.3 Schema — ActivityParticipants

For meetings and calls with multiple participants (internal or external):

New Table: `crm.ActivityParticipants`

Column	Type	Nullable	Default	Description
activity_participant_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
activity_id	BIGINT	NOT NULL	FK → Activities	Parent activity
participant_type	NVARCHAR(20)	NOT NULL		INTERNAL_USER, CUSTOMER_CONTACT, EXTERNAL
user_id	BIGINT	NULL	FK → auth.Users	For INTERNAL_USER type
contact_person_id	BIGINT	NULL	FK → SupplierContacts	For CUSTOMER_CONTACT type
external_name	NVARCHAR(200)	NULL		For EXTERNAL type (e.g. third-party consultant)
external_email	NVARCHAR(200)	NULL		
role	NVARCHAR(30)	NULL		ORGANIZER, REQUIRED, OPTIONAL, ATTENDEE

Index:

Index	Columns	Type
IX_ActivityParticipants_Activity	activity_id	Non-unique

---

11. C9 — Activity Documents

11.1 Overview

Every document shared with or received from the customer during the sales process is tracked with direction, context, and stage reference. Documents can be linked to a specific Activity (e.g., "brochure sent during the Oct 6 phone call") or to a Lead/Opportunity directly.

11.2 Schema — ActivityDocuments

New Table: `crm.ActivityDocuments`

Column	Type	Nullable	Default	Description
activity_document_id	BIGINT IDENTITY	NOT NULL	PK	Primary key
org_id	INT	NOT NULL	FK → tenants	Tenant discriminator
— Parent References —				
activity_id	BIGINT	NULL	FK → Activities	Linked activity (which conversation)
lead_id	BIGINT	NULL	FK → Leads	Linked lead
opportunity_id	BIGINT	NULL	FK → Opportunities	Linked opportunity
— Document Info —				
document_type	NVARCHAR(30)	NOT NULL	'OTHER'	BROCHURE, PRICE_LIST, PROPOSAL, SPECIFICATION, CONTRACT, CUSTOMER_PO, QUOTATION_PDF, PRESENTATION, IMAGE, MEETING_MINUTES, NDA, OTHER
document_name	NVARCHAR(200)	NOT NULL		Display name
file_id	BIGINT	NOT NULL	FK → core.FileStorage	Stored file reference
direction	NVARCHAR(30)	NOT NULL		SENT_TO_CUSTOMER, RECEIVED_FROM_CUSTOMER, INTERNAL
— Context —				

Column	Type	Nullable	Default	Description
pipeline_stage_id	INT	NULL	FK → PipelineStages	Stage when document was shared
notes	NVARCHAR(500)	NULL		
— Audit —				
shared_by_user_id	BIGINT	NOT NULL	FK → auth.Users	Who uploaded/shared
shared_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	When
created_at	DATETIME2	NOT NULL	SYSUTCDATETIME()	
is_deleted	BIT	NOT NULL	0	Soft delete

Indexes:

Index	Columns	Type
IX_ActivityDocuments_Activity	activity_id	Non-unique, filtered
IX_ActivityDocuments_Lead	lead_id	Non-unique, filtered
IX_ActivityDocuments_Opportunity	opportunity_id	Non-unique, filtered

---

12. C10 — Timeline View, Export & Reporting

12.1 Overview

The Timeline is a unified chronological view of ALL interactions on a Lead or Opportunity, aggregating data from multiple sources into a single scrollable feed.

12.2 Timeline Data Sources

The timeline for a Lead or Opportunity merges the following records:

Source	Displayed As	From Table
Activities (calls, meetings, emails, notes)	Activity cards with icon, subject, notes, outcome	crm.Activities
Documents shared/received	Document cards with filename, direction, type	crm.ActivityDocuments
Status changes	Status transition badges	Derived from Lead/Opportunity audit trail
Pipeline stage changes	Stage transition badges	Derived from Lead/Opportunity audit trail
Linked SCM documents	Reference cards (Inquiry created, Quotation sent, etc.)	demand.SaleInquiries, SaleQuotations, SaleOrders via FKs
Conversion event	Conversion card (Lead → Opportunity)	Lead.converted_at

12.3 Timeline API

Endpoint: `GET /api/crm/leads/{id}/timeline` **Endpoint:** `GET /api/crm/opportunities/{id}/timeline`

Response: Array of timeline entries sorted by date descending.

```json
```

```
{
  "timeline": [
    {
      "entry_type": "ACTIVITY",
      "activity_type": "PHONE_CALL",
      "date": "2026-10-05T14:30:00Z",
      "subject": "Initial contact with Ahmed Khan",
      "description": "Customer interested in 1.5-ton inverter AC...",
      "outcome": "FOLLOW_UP_NEEDED",
      "performed_by": { "user_id": 42, "name": "Salim Ahmed" },
      "contact_person": { "contact_id": 101, "name": "Ahmed Khan" },
      "duration_minutes": 15,
      "pipeline_stage": "Initial Contact",
      "documents": [],
      "follow_up": { "date": "2026-10-06", "type": "EMAIL_SENT", "notes": "Send product cat
    },
    {
      "entry_type": "DOCUMENT",
      "document_type": "BROCHURE",
      "date": "2026-10-06T09:15:00Z",
      "document_name": "AC-Product-Catalog-2026.pdf",
      "direction": "SENT_TO_CUSTOMER",
      "shared_by": { "user_id": 42, "name": "Salim Ahmed" },
      "pipeline_stage": "Initial Contact"
    },
    {
      "entry_type": "STATUS_CHANGE",
      "date": "2026-10-08T11:00:00Z",
      "from_status": "CONTACTED",
      "to_status": "QUALIFIED",
      "changed_by": { "user_id": 42, "name": "Salim Ahmed" }
    }
  ],
  "total_count": 12,
  "page": 1,
  "page_size": 25
}
```


12.4 Timeline Export — Excel

Endpoint: `GET /api/crm/leads/{id}/timeline/export?format=xlsx` **Endpoint:** `GET /api/crm/opportunities/{id}/timeline/export?format=xlsx`

Generates an Excel workbook with:

Sheet	Contents
Summary	Lead/Opportunity header info, account, contact, source, current status, total activities count
Timeline	One row per timeline entry: Date, Type, Subject, Description, Outcome, Performed By, Contact, Duration, Stage, Follow-up Date, Follow-up Notes
Documents	One row per document: Date, Name, Type, Direction, Shared By, Stage, Notes

12.5 Cross-Reference Panel

When viewing a Lead or Opportunity, the UI sidebar shows:

"Other Leads for this Account" — All leads (current and historical) for the same BusinessPartner (account_id), excluding the current record. Shows lead_number, topic, status, date.

"Other Leads for this Contact" — All leads where contact_person_id matches, AND account_id matches (data isolation — never show leads from other companies). Shows lead_number, topic, status, date.

Data Isolation Enforcement:

```sql
```

```sql
-- Other leads for same account
SELECT lead_number, topic, status, created_at
FROM crm.Leads
WHERE org_id = @OrgId
  AND account_id = @CurrentAccountId
  AND lead_id != @CurrentLeadId
  AND is_deleted = 0
ORDER BY created_at DESC;

-- Other leads for same contact at same account (NOT cross-company)
SELECT lead_number, topic, status, created_at
FROM crm.Leads
WHERE org_id = @OrgId
  AND contact_person_id = @CurrentContactPersonId
  AND account_id = @CurrentAccountId  -- ← CRITICAL: same company only
  AND lead_id != @CurrentLeadId
  AND is_deleted = 0
ORDER BY created_at DESC;
```

---

13. Database Migrations

Migration	Description	Objects
M1	Create `crm` schema	Schema
M2	Create crm.LeadSources	Table + seed data
M3	Create crm.Campaigns	Table + indexes
M4	Create crm.SalesTeams	Table + indexes
M5	Create crm.SalesTeamMembers	Table + indexes
M6	Create crm.PipelineStages	Table + seed data
M7	Create crm.Leads	Table + 12 indexes
M8	Create crm.Opportunities	Table + 8 indexes
M9	Create crm.Activities	Table + constraint + 7 indexes
M10	Create crm.ActivityParticipants	Table + index
M11	Create crm.ActivityDocuments	Table + 3 indexes
M12	Update core.ModuleCatalog — CRM is_available = 1	Update row
M13	Update core.ModuleFeatures — insert CRM features	Insert rows
M14	Alter demand.SaleOrders — add source_opportunity_id, update source_type constraint	Alter table

Migration Execution Order: M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8 → M9 → M10 → M11 → M12 → M13 → M14

---

14. API Endpoints

14.1 Lead Endpoints

Method	Route	Description	Permission
GET	/api/crm/leads	List leads with filtering, sorting, pagination	crm_view_leads
GET	/api/crm/leads/{id}	Get lead detail	crm_view_leads
POST	/api/crm/leads	Create lead	crm_create_lead
PUT	/api/crm/leads/{id}	Update lead	crm_edit_lead
PATCH	/api/crm/leads/{id}/assign	Assign lead to user/team	crm_assign_lead
PATCH	/api/crm/leads/{id}/status	Change lead status	crm_edit_lead
POST	/api/crm/leads/{id}/qualify	Qualify lead (CONTACTED → QUALIFIED)	crm_qualify_lead
POST	/api/crm/leads/{id}/convert	Convert lead to Opportunity (C6 flow)	crm_convert_lead
POST	/api/crm/leads/{id}/disqualify	Disqualify lead	crm_disqualify_lead
POST	/api/crm/leads/{id}/reopen	Re-open disqualified lead	crm_reopen_lead
DELETE	/api/crm/leads/{id}	Soft delete	crm_delete_lead
GET	/api/crm/leads/{id}/timeline	Get unified timeline	crm_view_leads
GET	/api/crm/leads/{id}/timeline/export	Export timeline (xlsx/pdf)	crm_export_timeline
GET	/api/crm/leads/{id}/cross-references	Other leads for same account/contact	crm_view_leads
GET	/api/crm/leads/{id}/duplicate-check	Check for potential duplicate leads	crm_view_leads
GET	/api/crm/leads/kanban	Leads grouped by pipeline stage (Kanban)	crm_view_leads

14.2 Opportunity Endpoints

Method	Route	Description	Permission
GET	/api/crm/opportunities	List opportunities	crm_view_opportunities
GET	/api/crm/opportunities/{id}	Get opportunity detail	crm_view_opportunities
PUT	/api/crm/opportunities/{id}	Update opportunity	crm_edit_opportunity
PATCH	/api/crm/opportunities/{id}/stage	Change pipeline stage	crm_edit_opportunity
POST	/api/crm/opportunities/{id}/won	Mark as Won	crm_close_opportunity
POST	/api/crm/opportunities/{id}/lost	Mark as Lost (requires reason)	crm_close_opportunity
POST	/api/crm/opportunities/{id}/reopen	Re-open lost opportunity	crm_reopen_opportunity
POST	/api/crm/opportunities/{id}/create-inquiry	Create Sale Inquiry from Opportunity	crm_edit_opportunity + demand_create_inquiry
POST	/api/crm/opportunities/{id}/create-quotation	Create Sale Quotation from Opportunity	crm_edit_opportunity + demand_create_quotation
POST	/api/crm/opportunities/{id}/create-order	Create Sale Order from Opportunity	crm_edit_opportunity + demand_create_order
GET	/api/crm/opportunities/{id}/timeline	Get unified timeline	crm_view_opportunities
GET	/api/crm/opportunities/{id}/timeline/export	Export timeline (xlsx/pdf)	crm_export_timeline

Method	Route	Description	Permission
GET	/api/crm/opportunities/kanban	Opportunities grouped by pipeline stage	crm_view_opportunities

14.3 Activity Endpoints

Method	Route	Description	Permission
GET	/api/crm/activities	List activities (filterable by lead, opportunity, user, type, date range)	crm_view_activities
GET	/api/crm/activities/{id}	Get activity detail	crm_view_activities
POST	/api/crm/activities	Create activity	crm_create_activity
PUT	/api/crm/activities/{id}	Update activity	crm_edit_activity
PATCH	/api/crm/activities/{id}/complete	Mark activity complete	crm_edit_activity
DELETE	/api/crm/activities/{id}	Soft delete	crm_delete_activity
POST	/api/crm/activities/{id}/documents	Attach document to activity	crm_create_activity
GET	/api/crm/activities/pending	Get pending follow-ups and tasks	crm_view_activities

14.4 Campaign Endpoints

Method	Route	Description	Permission
GET	/api/crm/campaigns	List campaigns	crm_view_campaigns
GET	/api/crm/campaigns/{id}	Get campaign detail with lead count	crm_view_campaigns
POST	/api/crm/campaigns	Create campaign	crm_manage_campaigns
PUT	/api/crm/campaigns/{id}	Update campaign	crm_manage_campaigns
DELETE	/api/crm/campaigns/{id}	Soft delete	crm_manage_campaigns

14.5 Configuration Endpoints

Method	Route	Description	Permission
GET	/api/crm/lead-sources	List lead sources	crm_view_leads
POST	/api/crm/lead-sources	Create custom lead source	crm_admin
PUT	/api/crm/lead-sources/{id}	Update lead source	crm_admin
GET	/api/crm/pipeline-stages	List pipeline stages	crm_view_leads
POST	/api/crm/pipeline-stages	Create pipeline stage	crm_admin
PUT	/api/crm/pipeline-stages/{id}	Update pipeline stage (name, sequence, probability)	crm_admin
DELETE	/api/crm/pipeline-stages/{id}	Deactivate stage (only if no leads in it)	crm_admin
GET	/api/crm/sales-teams	List sales teams	crm_view_leads
POST	/api/crm/sales-teams	Create team	crm_admin
PUT	/api/crm/sales-teams/{id}	Update team	crm_admin
POST	/api/crm/sales-teams/{id}/members	Add member	crm_admin
DELETE	/api/crm/sales-teams/{id}/members/{userId}	Remove member	crm_admin

14.6 Public Lead Ingestion API (Foundation for Addendum 42)

This endpoint is designed for external lead creation — website forms, third-party integrations, and future embeddable widgets. It uses API key authentication instead of user sessions.

Method	Route	Description	Auth
POST	/api/public/crm/leads	Create lead from external source	API Key (X-Api-Key header)

Request body:

```
json
```

```
{
  "first_name": "Ahmed",
  "last_name": "Khan",
  "company_name": "PakElectro",
  "email": "ahmed@pakelec.com",
  "phone": "+92-300-1234567",
  "topic": "Interested in inverter AC",
  "description": "Saw your Facebook ad for summer AC sale",
  "source_code": "FACEBOOK",
  "campaign_code": "SUMMER-AC-2026",
  "source_medium": "social",
  "source_reference": "fb-ad-12345",
  "external_source_system": null,
  "external_source_id": null
}
```

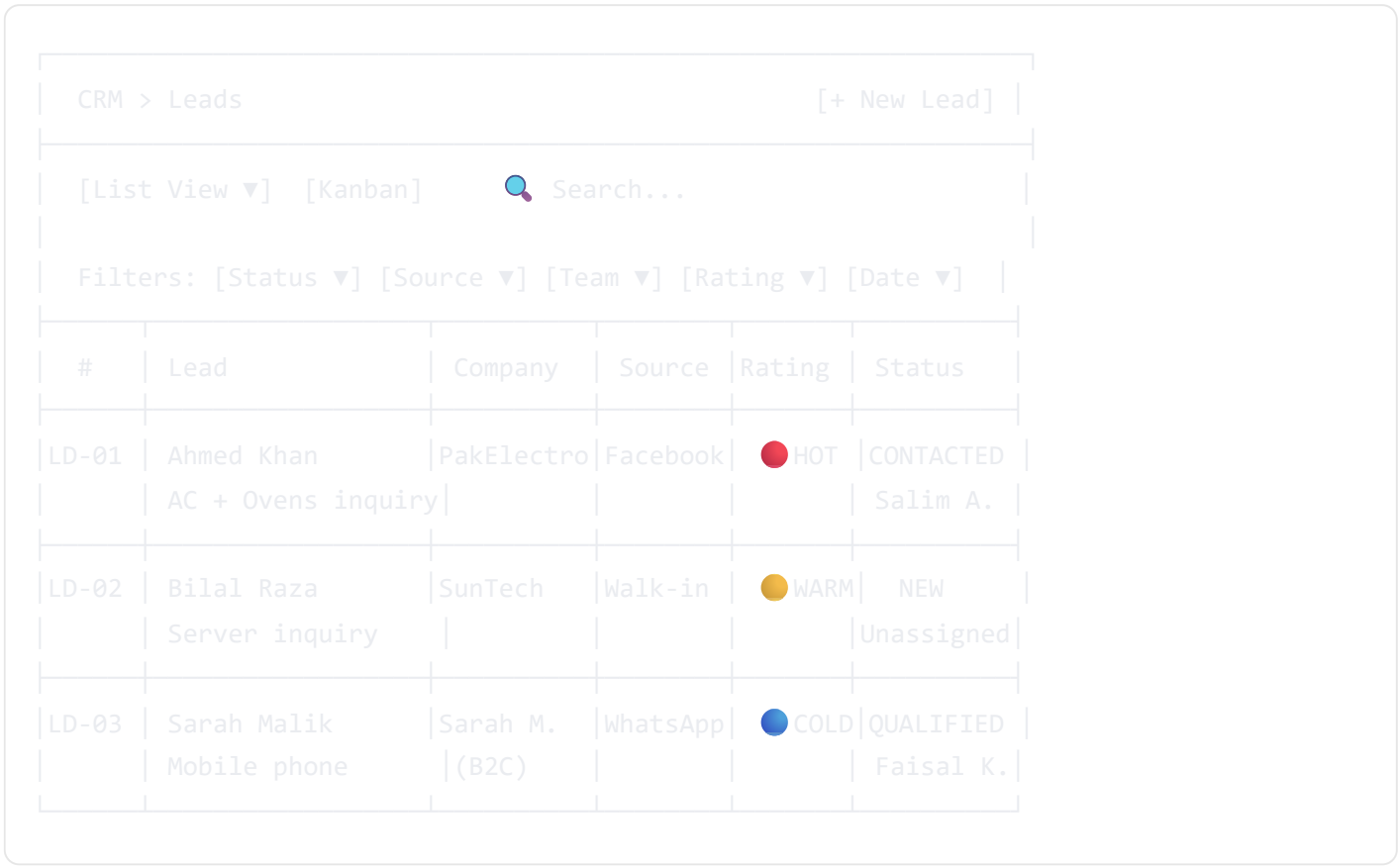
Response: Returns lead_number and lead_id. The lead is created with status = NEW and no assignment. Assignment rules (Addendum 42) will auto-assign; until then, a manager assigns manually.

Rate Limiting: 100 requests per minute per API key. Configurable per org.

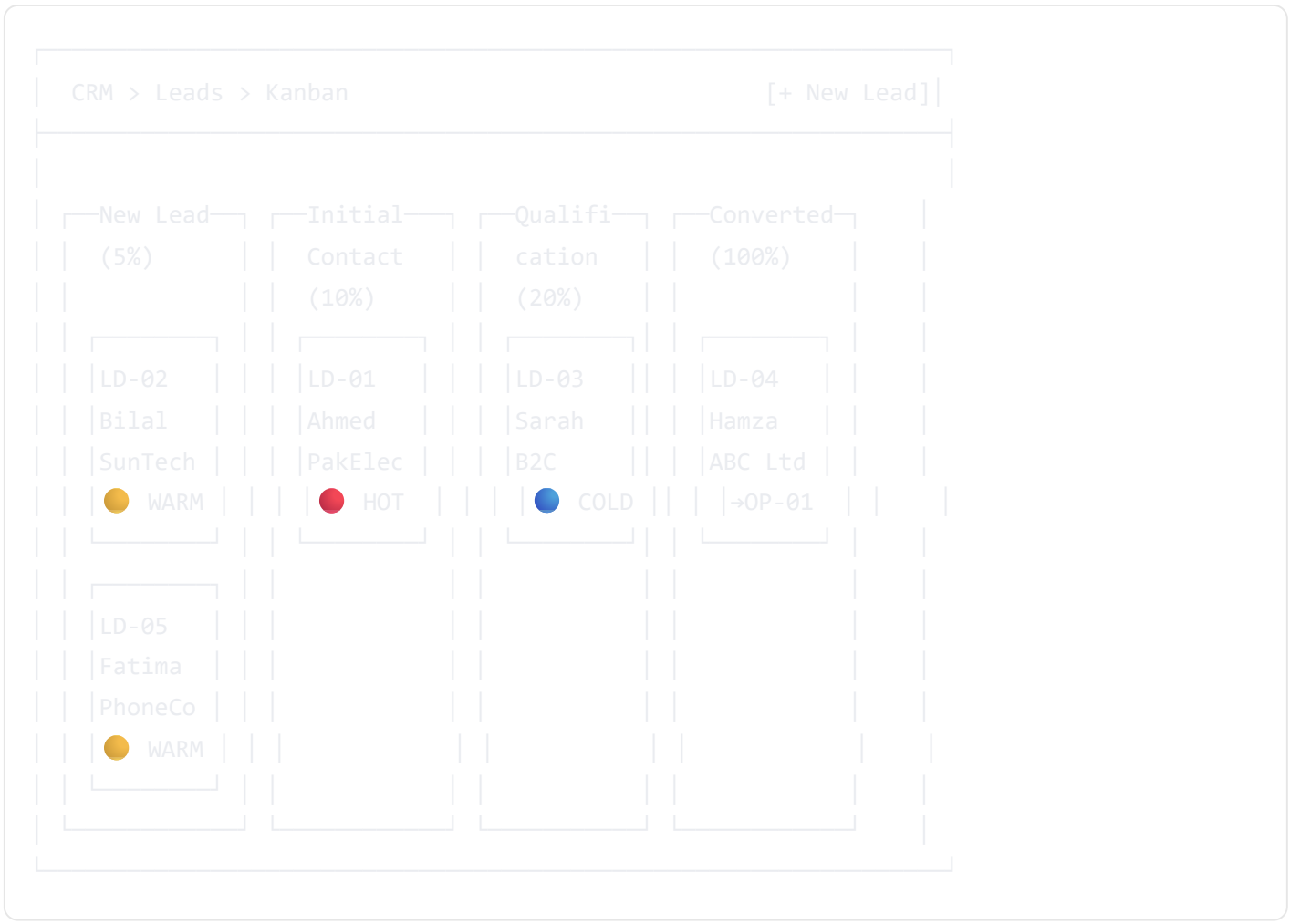
---

15. UI Wireframes & Specifications

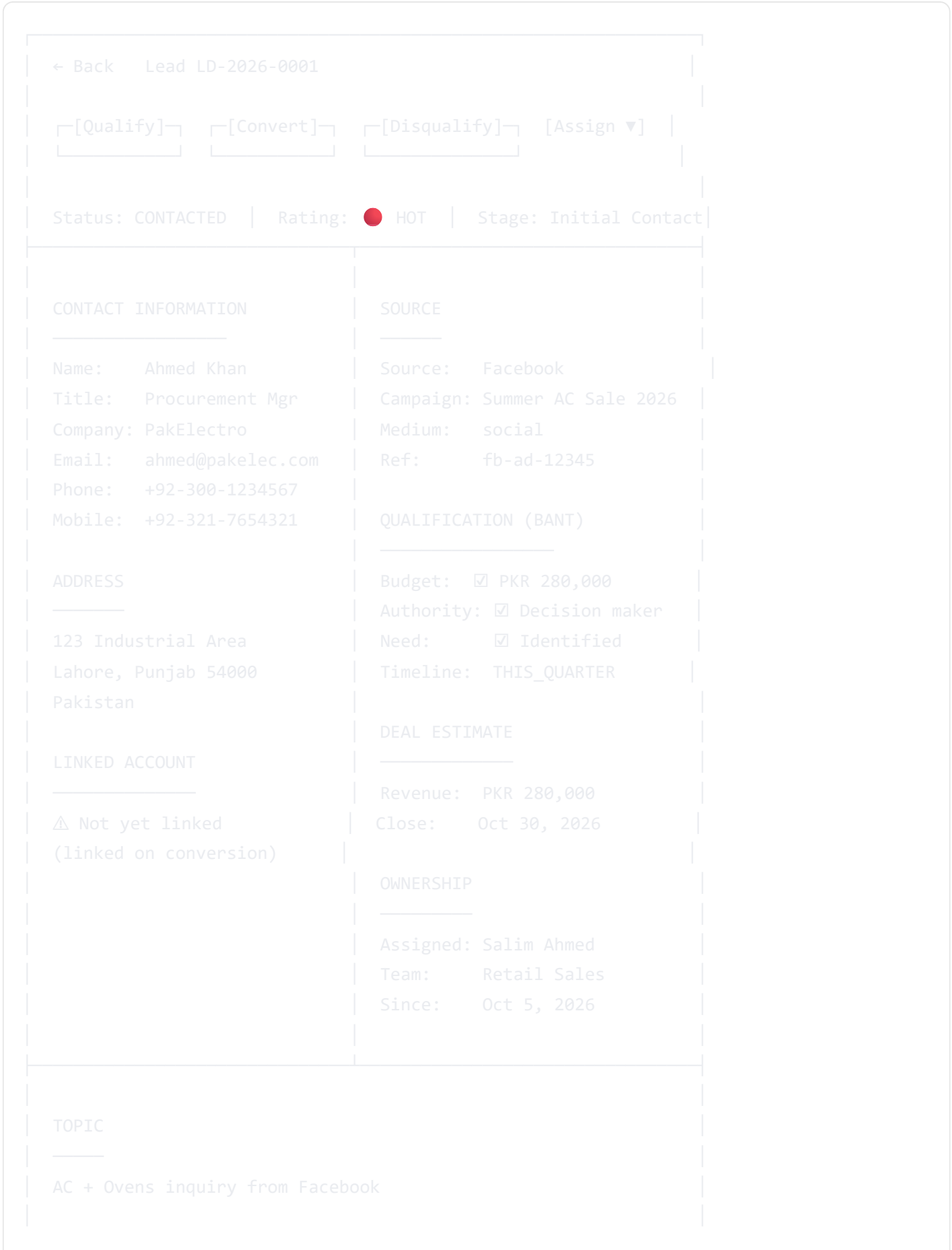
15.1 Lead List View



15.2 Lead Kanban View



15.3 Lead Detail View



DESCRIPTION

Customer responded to Summer AC Sale Facebook ad. Interested in 1 inverter AC (1.5 ton), 2 ovens, and possibly a mobile phone. Budget approximately PKR 280,000.

TIMELINE

[Export ▼]

[+ Call] [+ Meeting] [+ Note] [+ Task]
[+ Email] [+ Document]

- Oct 8 — MEETING (45 min)
  Salim + Faisal → Ahmed Khan
  PakElectro office
  "Customer confirmed specs. Wants Haier 1.5-ton inverter, 2x cooking range... Requested formal quotation by Oct 10."
  Outcome: ✅ Successful
  📎 Meeting-Minutes-Oct8.docx (Internal)
  Next: Create Quotation by Oct 10

- Oct 6 — DOCUMENT SHARED
  → Customer: AC-Product-Catalog-2026.pdf
  → Customer: Oven-Range-Brochure.pdf
  By: Salim Ahmed

- Oct 5 — PHONE CALL (15 min)
  Salim → Ahmed Khan
  "Customer interested in 1.5-ton..."
  Outcome: Follow-up needed
  Next: Send catalog by Oct 6

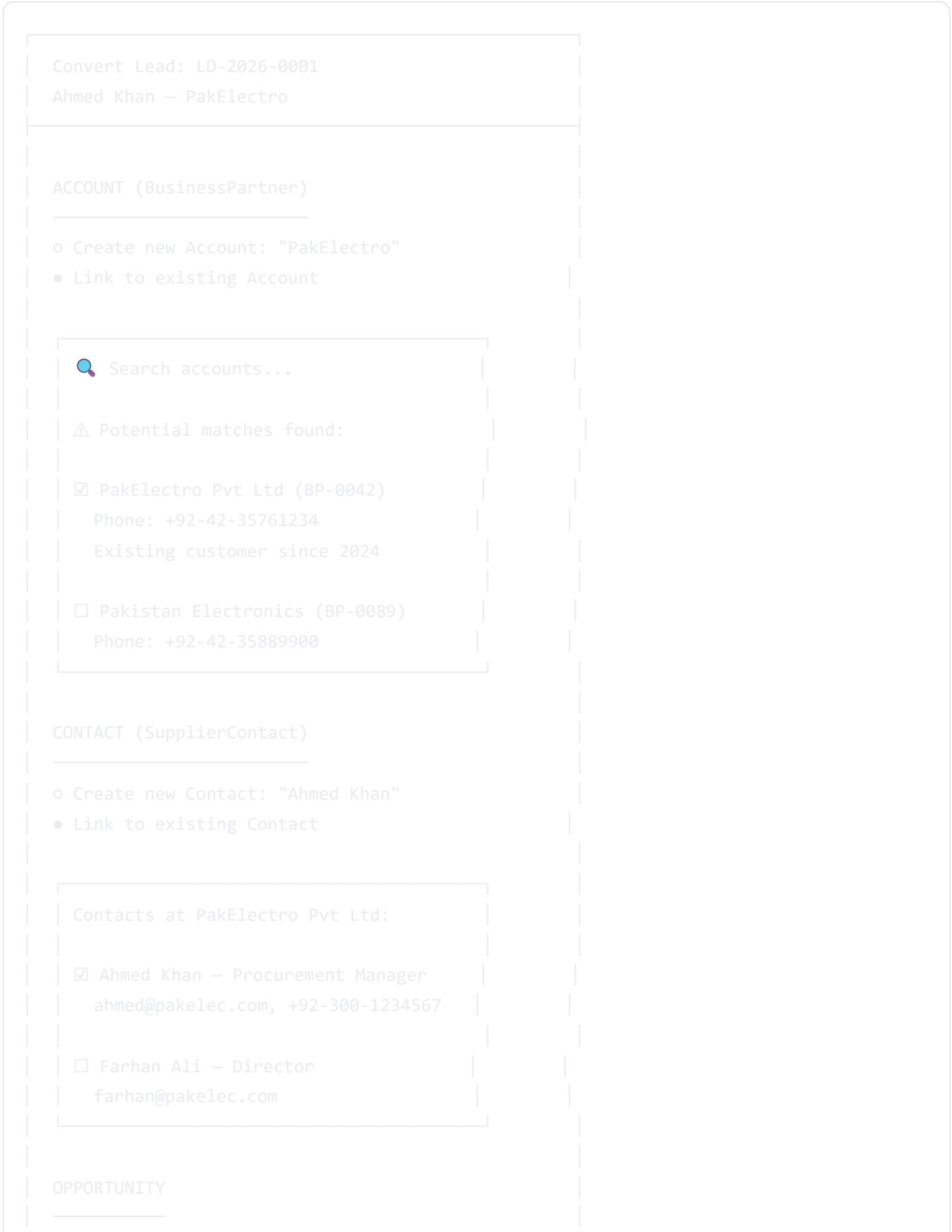
- Oct 5 — LEAD CREATED
  Source: Facebook / Summer AC Sale 2026
  Assigned to: Salim Ahmed

CROSS-REF

Other leads for PakElec:
(none)

Other leads for Ahmed:
(none)

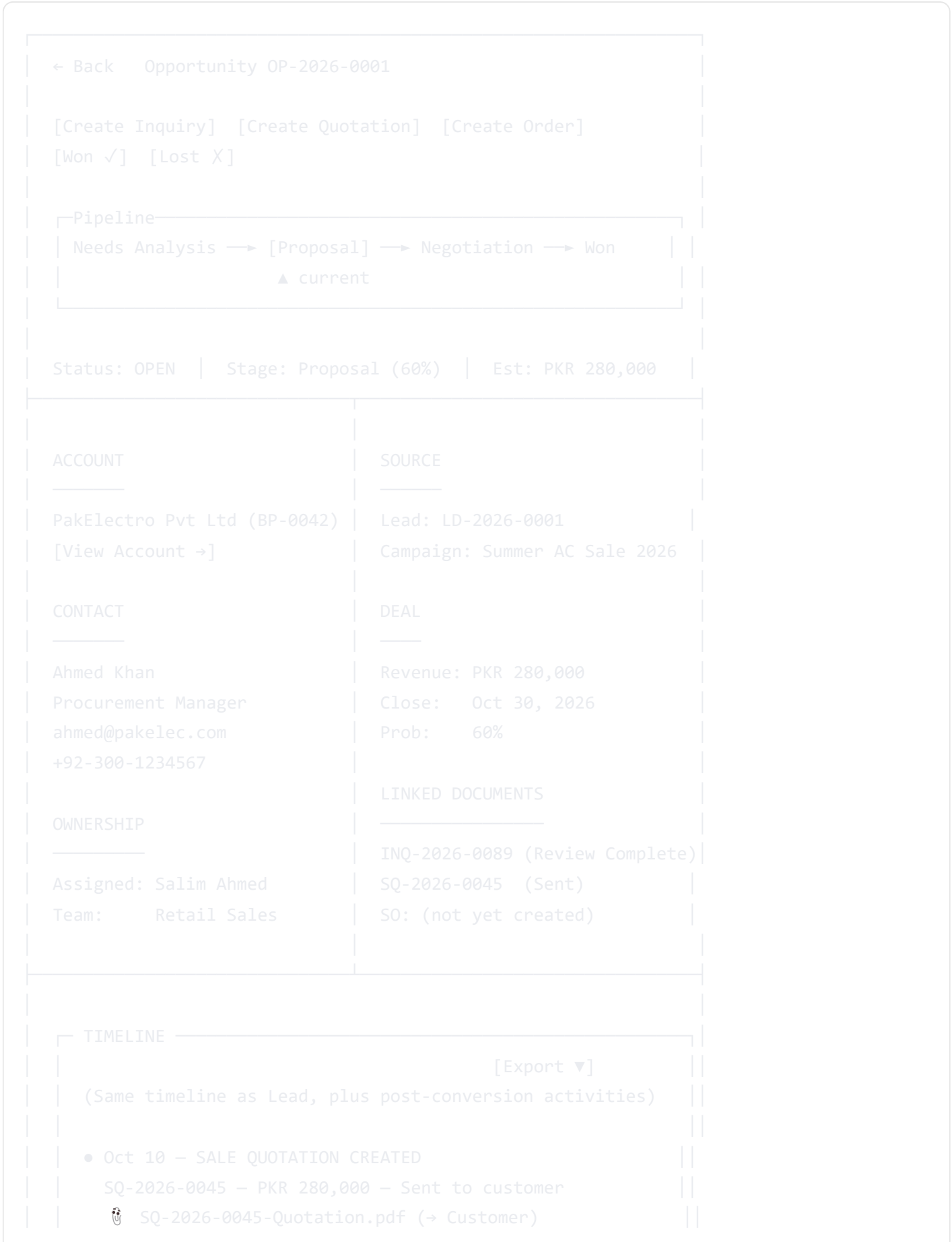
15.4 Lead Conversion Dialog



Topic: AC + Ovens inquiry from Facebook
Revenue: PKR 280,000
Close by: Oct 30, 2026
Assign to: Salim Ahmed

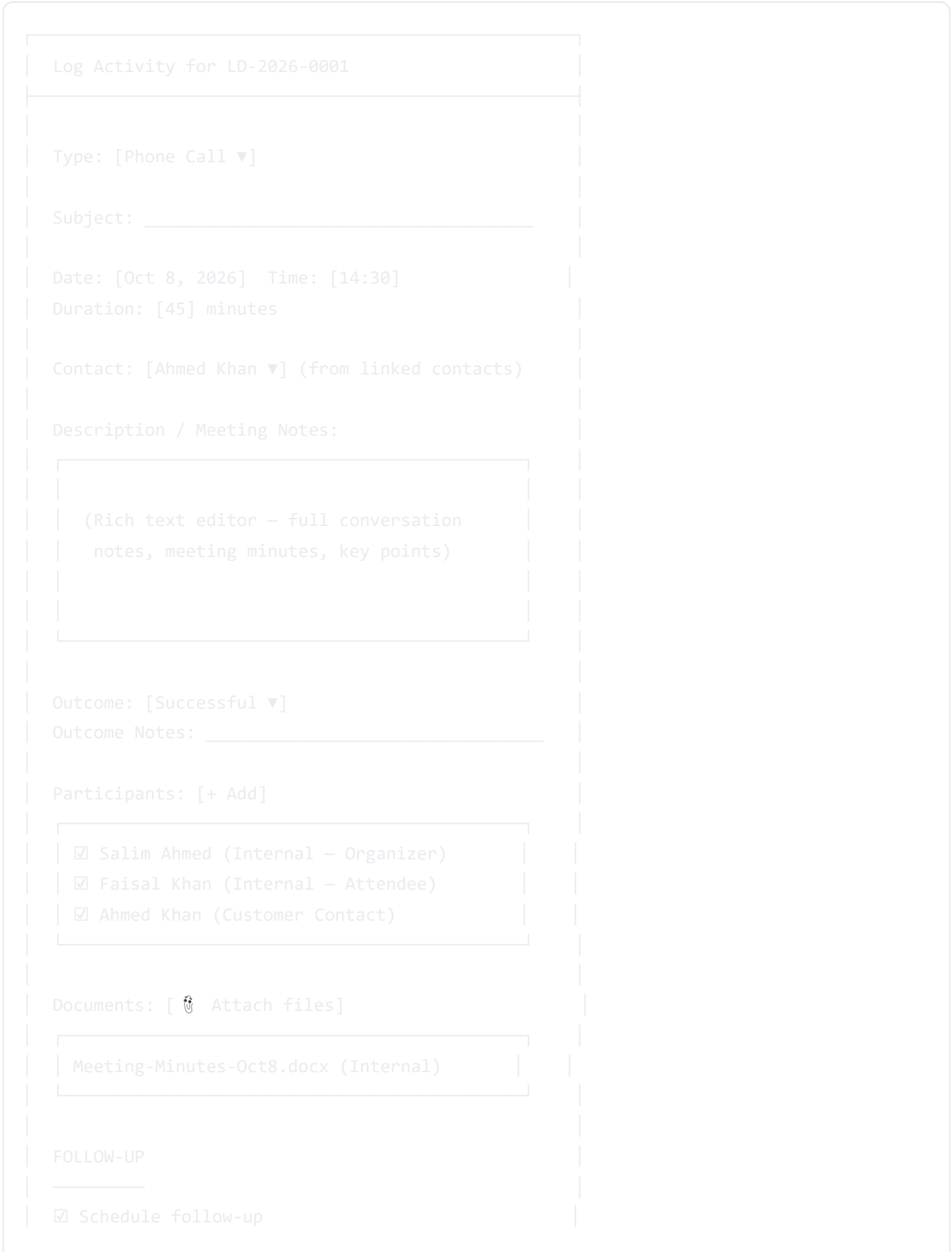
Convert Cancel

15.5 Opportunity Detail View



• Oct 9 — SALE INQUIRY CREATED
INQ-2026-0089 — 3 lines, all CAN_SUPPLY
• Oct 8 — LEAD CONVERTED TO OPPORTUNITY
Account: PakElectro Pvt Ltd (linked existing)
Contact: Ahmed Khan (linked existing)
—— Activities from Lead LD-2026-0001 ——
• Oct 8 — MEETING (45 min)
... (carried from lead)

15.6 Activity Creation Form



Date: [Oct 10, 2026]
Type: [Task ▼]
Notes: Create and send formal quotation

Save Cancel

16. Business Rules

16.1 Lead Rules

#	Rule	Enforcement
BR-01	Lead number (LD-{YYYY}-{NNNN}) is auto-generated per org, per year, sequential	Backend — NumberingService
BR-02	`first_name`, `last_name`, and `company_name` are required on creation	Backend validation
BR-03	For B2C individual leads, `company_name` must be set to the person's full name	UI guidance + backend default
BR-04	Lead status transitions must follow the state machine (C4, section 6.3)	Backend — state machine service
BR-05	Only QUALIFIED leads can be converted to Opportunities	Backend — ConvertAsync validates
BR-06	A CONVERTED lead is terminal — cannot be edited, re-opened, or deleted	Backend
BR-07	A DISQUALIFIED lead can only be re-opened by users with `crm_reopen_lead` permission	Backend + RBAC
BR-08	Lead-to-CONTACTED transition requires at least one logged Activity	Backend validation
BR-09	Duplicate detection runs on create and on convert (email, phone, company_name match)	Backend — advisory, not blocking
BR-10	Lead assignment updates `assigned_at` and `previous_owner_user_id`	Backend

16.2 Opportunity Rules

#	Rule	Enforcement
BR-11	Opportunities are ONLY created via Lead conversion — no standalone creation	Backend — no direct create endpoint
BR-12	WON status requires a linked Sale Order with status ⩾ CONFIRMED	Backend validation
BR-13	LOST status requires `win_loss_reason`	Backend validation
BR-14	LOST opportunities can be re-opened with `crm_reopen_opportunity` permission	Backend + RBAC
BR-15	When an Opportunity creates a Sale Inquiry, `customer_id` on the Inquiry is auto-set from `account_id`	Backend
BR-16	`actual_revenue` is computed from the linked Sale Order's grand total on WON	Backend

16.3 Activity Rules

#	Rule	Enforcement
BR-17	Every Activity must be linked to at least one of: lead_id, opportunity_id	Database constraint
BR-18	Activities on a Lead are visible on its converted Opportunity (via source_lead_id join)	Backend — timeline query
BR-19	Activity `description` supports rich text (stored as HTML) for detailed meeting minutes	Frontend + backend
BR-20	Follow-up date creates a pending task visible in the salesperson's activity dashboard	Frontend + query on follow_up_date

16.4 Data Isolation Rules

#	Rule	Enforcement
BR-21	org_id tenant filter applies to ALL CRM entities via EF Core HasQueryFilter	EF Core global filter
BR-22	Cross-reference queries for same contact MUST filter by account_id	Backend — hardcoded in query
BR-23	Same person at multiple companies has separate SupplierContact rows; CRM isolation is at Account level	Architecture

17. Test Scenarios

17.1 Lead Lifecycle

#	Scenario	Expected Result
TS-01	Create lead with all required fields	Lead created with status NEW, lead_number generated
TS-02	Create B2C lead (no company)	company_name auto-set to full_name
TS-03	Attempt status change NEW → QUALIFIED (no activities)	Rejected — must be CONTACTED first
TS-04	Log a phone call on NEW lead	Lead status transitions to CONTACTED
TS-05	Qualify CONTACTED lead	Status changes to QUALIFIED
TS-06	Convert QUALIFIED lead (new account + new contact)	BP created, Contact created, Opportunity created, Lead = CONVERTED
TS-07	Convert QUALIFIED lead (link existing account + existing contact)	No new BP/Contact; Opportunity created; Lead = CONVERTED
TS-08	Convert lead with duplicate BP match	Duplicate warning shown; user chooses link or create
TS-09	Disqualify lead	Status = DISQUALIFIED, reason recorded
TS-10	Re-open disqualified lead (with permission)	Status = NEW
TS-11	Re-open disqualified lead (without permission)	Rejected — 403
TS-12	Edit CONVERTED lead	Rejected — terminal state

17.2 Opportunity Lifecycle

#	Scenario	Expected Result
TS-13	Create Inquiry from Opportunity	Inquiry created with customer_id = opportunity.account_id, source linked
TS-14	Create Quotation from Opportunity	Quotation created with customer_id = opportunity.account_id, source linked
TS-15	Mark Opportunity as Won (SO confirmed)	Status = WON, actual_revenue set, actual_close_date set
TS-16	Mark Opportunity as Won (no SO)	Rejected — requires linked confirmed SO
TS-17	Mark Opportunity as Lost (with reason)	Status = LOST, reason recorded
TS-18	Mark Opportunity as Lost (no reason)	Rejected — reason required

17.3 Timeline & Activities

#	Scenario	Expected Result
TS-19	Get timeline for lead with activities + docs + status changes	All entries merged, sorted by date desc
TS-20	Get timeline for opportunity (includes lead activities)	Lead activities shown below conversion line
TS-21	Export timeline as Excel	Workbook generated with Summary, Timeline, Documents sheets
TS-22	Attach document to activity	Document linked to activity and visible in timeline

17.4 Data Isolation

#	Scenario	Expected Result
TS-23	Ahmed Khan at Company A — view cross-references	Only Company A leads shown
TS-24	Ahmed Khan at Company B — view cross-references	Only Company B leads shown, Company A data invisible
TS-25	Same org, different tenant	Tenant filter isolates all data

---

18. Development Phases

Phase 1 — Core CRM (This Addendum)

Priority	Component	Estimated Effort
P1	M1–M6: Schema + lookups + seed data	2 days
P1	M7: Leads table + CRUD API + List/Detail UI	5 days
P1	Lead state machine + status transitions	2 days
P1	M9–M10: Activities + CRUD API + Timeline UI	5 days
P1	M11: Activity Documents	2 days
P1	M8: Opportunities table + CRUD API + Detail UI	3 days
P1	C6: Lead Conversion Service + Conversion Dialog UI	4 days
P1	M14: SaleOrder extension (source_opportunity_id)	1 day
P1	Kanban view (Lead + Opportunity)	3 days
P1	Timeline export (Excel)	2 days
P1	Cross-reference panel	1 day
P1	RBAC permissions setup	1 day
	Phase 1 Total	**~31 days**

Phase 2 — Assignment & Automation (Addendum 42)

Component	Description
Lead Assignment Rules	Round-robin, territory, campaign-based, workload-based auto-assignment
SLA Enforcement	Auto-escalation if lead not contacted within X hours
Lead Scoring	Rule-based scoring engine
Follow-up Reminders	Hangfire jobs for pending follow-ups
Duplicate Detection Rules	Configurable matching rules beyond simple field comparison
Pipeline Analytics Dashboard	Win rate, conversion funnel, average deal time

Phase 3 — Integration & Widgets (Addendum 42 or 43)

Component	Description
Embeddable Lead Widget	JavaScript snippet + widget configuration UI
Odoo Sync Connector	Import leads from Odoo via API
SAP Sync Connector	Import leads from SAP via API
Dynamics Sync Connector	Import leads from Dynamics via API
Webhook Outbound	Push lead events to external systems
Facebook Lead Ads Integration	Auto-create leads from Facebook Lead Ads
WhatsApp Business Integration	Auto-create leads from WhatsApp messages
Timeline PDF Export	Formatted PDF report generation

Appendix A — RBAC Permission Matrix

Permission Code	Description	Default Roles
crm_view_leads	View leads list and detail	Sales Rep, Sales Manager, Admin
crm_create_lead	Create new leads	Sales Rep, Sales Manager, Admin
crm_edit_lead	Edit lead details	Sales Rep (own), Sales Manager, Admin
crm_assign_lead	Assign/reassign leads	Sales Manager, Admin
crm_qualify_lead	Qualify a lead	Sales Rep, Sales Manager, Admin
crm_convert_lead	Convert lead to opportunity	Sales Rep, Sales Manager, Admin
crm_disqualify_lead	Disqualify a lead	Sales Rep, Sales Manager, Admin
crm_reopen_lead	Re-open a disqualified lead	Sales Manager, Admin
crm_delete_lead	Soft-delete a lead	Sales Manager, Admin
crm_view_opportunities	View opportunities	Sales Rep, Sales Manager, Admin
crm_edit_opportunity	Edit opportunity details	Sales Rep (own), Sales Manager, Admin
crm_close_opportunity	Mark opportunity as Won/Lost	Sales Rep (own), Sales Manager, Admin
crm_reopen_opportunity	Re-open a lost opportunity	Sales Manager, Admin
crm_view_activities	View activities and timeline	Sales Rep, Sales Manager, Admin
crm_create_activity	Log new activities	Sales Rep, Sales Manager, Admin

Permission Code	Description	Default Roles
crm_edit_activity	Edit activities	Sales Rep (own), Sales Manager, Admin
crm_delete_activity	Delete activities	Sales Manager, Admin
crm_export_timeline	Export timeline to Excel/PDF	Sales Rep, Sales Manager, Admin
crm_view_campaigns	View campaigns	Sales Rep, Sales Manager, Admin
crm_manage_campaigns	Create/edit/delete campaigns	Sales Manager, Admin
crm_admin	CRM configuration (pipeline stages, lead sources, teams)	Admin

Appendix B — MediatR Events

Event	Published When	Payload
LeadCreatedEvent	New lead created	LeadId, OrgId, LeadSourceId, CampaignId
LeadAssignedEvent	Lead assigned/reassigned	LeadId, OrgId, NewUserId, PreviousUserId
LeadStatusChangedEvent	Lead status transition	LeadId, OrgId, OldStatus, NewStatus, Reason
LeadQualifiedEvent	Lead qualified	LeadId, OrgId
LeadConvertedEvent	Lead converted to Opportunity	LeadId, OpportunityId, AccountId, ContactId, OrgId
LeadDisqualifiedEvent	Lead disqualified	LeadId, OrgId, Reason
OpportunityCreatedEvent	Opportunity created (via conversion)	OpportunityId, LeadId, AccountId, OrgId
OpportunityStageChangedEvent	Pipeline stage changed	OpportunityId, OrgId, OldStageId, NewStageId
OpportunityWonEvent	Opportunity marked won	OpportunityId, OrgId, SaleOrderId, ActualRevenue
OpportunityLostEvent	Opportunity marked lost	OpportunityId, OrgId, Reason, Competitor
ActivityCreatedEvent	Activity logged	ActivityId, LeadId, OpportunityId, ActivityType, OrgId

These events enable future cross-module integrations (notifications, dashboards, analytics) without tight coupling.

Appendix C — Dynamics 365 Field Mapping Reference

For future data sync (Addendum 42), this table maps Dynamics fields to SMS fields:

Dynamics Field	SMS Field	Entity
lead.topic	Leads.topic	crm.Leads
lead.firstname	Leads.first_name	crm.Leads
lead.lastname	Leads.last_name	crm.Leads
lead.jobtitle	Leads.job_title	crm.Leads
lead.emailaddress1	Leads.email	crm.Leads
lead.telephone1	Leads.phone	crm.Leads
lead.mobilephone	Leads.mobile_phone	crm.Leads
lead.companyname	Leads.company_name	crm.Leads
lead.websiteurl	Leads.website	crm.Leads
lead.budgetamount	Leads.budget_amount	crm.Leads
lead.decisionmaker	Leads.authority_confirmed	crm.Leads
lead.need	Leads.need_identified	crm.Leads
lead.purchasetimeframe	Leads.purchase_timeframe	crm.Leads
lead.leadqualitycode	Leads.rating	crm.Leads
lead.estimatedvalue	Leads.estimated_revenue	crm.Leads
lead.estimatedclosedate	Leads.estimated_close_date	crm.Leads
lead.leadsourcecode	Leads.lead_source_id	crm.Leads (via LeadSources lookup)
lead.campaignid	Leads.campaign_id	crm.Leads
lead.statuscode	Leads.status	crm.Leads
opportunity.name	Opportunities.topic	crm.Opportunities
opportunity.estimatedvalue	Opportunities.estimated_revenue	crm.Opportunities
opportunity.estimatedclosedate	Opportunities.estimated_close_date	crm.Opportunities

Dynamics Field	SMS Field	Entity
opportunity.stepname	Opportunities.pipeline_stage_id	crm.Opportunities (via PipelineStages)
opportunity.statuscode	Opportunities.status	crm.Opportunities
activitypointer.subject	Activities.subject	crm.Activities
activitypointer.description	Activities.description	crm.Activities
activitypointer.activitytypecode	Activities.activity_type	crm.Activities
activitypointer.scheduledstart	Activities.scheduled_date	crm.Activities
activitypointer.actualdurationminutes	Activities.duration_minutes	crm.Activities

End of Addendum 41